

Adverse Selection - Hidden Information and Screening

Part II

Jonas Maibom
Aarhus University

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1 Monopolistic screening problem

Consider a transaction between a buyer and a seller, where the seller does not know perfectly how much the buyer is willing to pay for a good. Suppose, in addition, that the seller sets the terms of the contract (the seller is the Principal, the buyer is the agent). The **type** θ of the consumer is either low or high, i.e. $\theta_i \in \{\theta_L, \theta_H\}$ with $\theta_H > \theta_L$. The buyer is of type θ_L with probability β . The probability β can also be interpreted as the proportion of consumers of type θ_L . The buyer's preferences has the following form:

$$u(q, T, \theta) = \theta_i v(q) - T \quad (1)$$

with $v(0) = 0$, $v'(q) > 0$, and $v''(q) < 0$ for all q . **The characteristics θ_i are private information to the buyer**, the seller only knows the distribution of θ .

The seller's profit from selling q units of the good is

$$\pi = T - cq \quad (2)$$

where $c \geq 0$ is the unit cost of production.

What is the profit-maximizing pair (T, q) that the seller will be able to induce the buyer to choose?

1.1 The first-best outcome - Perfect information

Suppose here that information is perfect: the seller perfectly identifies each consumer θ_i and can offer him a specific contract (T_i, q_i) , $i = L, H$. This is known as perfect price discrimination.

The seller tries to maximize her profits subject to inducing the buyer to accept the proposed contract. Assume the buyer's reservation utility is $\underline{u}$ if he

does not take the seller's offer. The seller's problem is thus, for every $i = L, H$,

$$\max_{T_i, q_i} T_i - cq_i \quad \text{subject to } \mathbb{R}_i : \theta_i v(q_i) - T_i \geq \underline{u}. \quad (3)$$

The solution to this problem will be $(T_i^{FB}, q_i^{FB})_{i=L,H}$ such that

$$\theta_i v'(q_i^{FB}) = c \quad (4)$$

and

$$\theta_i v(q_i^{FB}) = T_i^{FB} - \underline{u}. \quad (5)$$

At the first-best outcome, the quantity sold is such that marginal utility equals marginal cost. The total transfer is set so that the seller appropriates the full surplus, and leaves no rent to the buyer above his reservation utility.

Without adverse selection (under the first best), the total profit of the seller is

$$\beta(T_L^{FB} - cq_L^{FB}) + (1 - \beta)(T_H^{FB} - cq_H^{FB}) \quad (6)$$

The optimal contract can be implemented by type-specific two-part tariffs, where the buyer is allowed to buy as much as he wants of the good at unit price c , provided that he pays a type-specific fixed fee equal to $\theta_i v(q_i^{FB}) - cq_i^{FB}$. For an illustration of the First-Best optimum see notes from class.

1.2 Adverse selection, linear pricing, and simple two-part tariffs

Now, consider that information is asymmetric, and that the seller cannot identify correctly the type of a buyer. If she still offers the two contracts designed for the first-best solution, the type- θ_H buyer will actually choose the contract (q_L^{FB}, T_L^{FB}) . Indeed, since $\theta_H > \theta_L$ and $T_L^{FB} < T_H^{FB}$, $\theta_H v(q_L^{FB}) - T_L^{FB} > \underline{u} = \theta_L v(q_L^{FB}) - T_L^{FB}$.

If the seller cannot observe the type of the agent, she must offer the same contracts to everybody. The contract set is potentially large, since it consists of the set of functions $T(q)$. We will first study two simple contracts of this form, linear pricing $T(q) = Pq$, and simple two-part tariffs $T(q) = F + Pq$.

1.2.1 Linear pricing

In this case, the seller only specifies a unit price P and not a quantity q . Given this contract (the unit price P), the buyer chooses q_i that maximizes $\theta_i v(q_i) - Pq_i$ where $i = L, H$. From the FOC, $\theta_i v'(q_i) = P$, we can derive the demand functions of each type: $q_i = D_i(P)$.

The buyer's net surplus can be written as follows:

$$S_i(P) = \theta_i v[D_i(P)] - PD_i(P). \quad (7)$$

Let

$$\begin{aligned} D(P) &\equiv \beta D_L(P) + (1 - \beta) D_H(P), \\ S(P) &\equiv \beta S_L(P) + (1 - \beta) S_H(P). \end{aligned}$$

With linear pricing, the seller's problem is a monopoly pricing problem, where the seller chooses P^m to solve

$$\max_P (P - c)D(P)$$

and the monopoly price is given by

$$P^m = c - \frac{D(P)}{D'(P)}.$$

In this solution, buyers keep a positive rent ($S(P^m) > 0$), and consumption is inefficiently low, since $\theta_i v'(q_i^M) = P^m > c$ implies $q_i^M < q^{FB}$. Note that depending on the values of β , θ_L and θ_H , it may be optimal for the seller to serve only the buyers of type θ_H . Note though, that consumption of q would still be inefficient compared to a two part tariff case under the first best (see discussion from slides under the first best).

1.2.2 Single two-part tariff

Suppose it is not the case that the seller wants to cater one type only. Can the seller then do better by offering nonlinear contracts? She might, if buyers cannot trade goods in a secondary market. If arbitrage is cost-less, then only linear pricing is possible, because buyers would buy at a minimum average price and then resell in the secondary market.

Here, we consider only one buyer and interpret β as a probability measure. There are no arbitrage opportunities open to the buyer. Therefore, a single two-part tariff is (F, P^F) , where P^F is the unit price and F the fixed fee which will allow P to improve upon linear pricing for the seller.

First, note that for any given price P , the maximum fixed fee the seller will set is $F = S_L(P)$ (this is the maximum fixed fee the buyer of type θ_L will accept to pay). A buyer of type θ_H will always decide to buy a positive quantity q when charged a two-part tariff $T(q) = S_L(P) + Pq$, since $\theta_H > \theta_L$ and thus $\theta_H v(q) - Pq > \theta_L v(q) - Pq = S_L(P)$.

If the seller decides to serve both types of consumers, and therefore sets $F = S_L(P)$, she also chooses P^F to solve

$$\max_P S_L(P) + (P - c)D(P). \quad (8)$$

The solution is given by

$$S'_L(P^F) + D(P^F) + (P^F - c)D'(P^F) = 0 \Rightarrow P^F = c - \left(\frac{D(P^F) + S'_L(P^F)}{D'(P^F)} \right). \quad (9)$$

By the envelope theorem, $S'_L(P) = -D_L(P)$, such that $D(P^F) + S'_L(P^F) > 0$. Since $D'(P) < 0$, $P^F > c$: the first-best outcome cannot be achieved and under-consumption remains.

Nevertheless, the seller prefers this two-part tariff to a linear contract, since she can always increase her profit by setting $F = S_L(P^m)$. We can also observe that $P^m > P^F > c$.

An important conclusion of the optimal single two-part tariff is that the θ_H -type buyer strictly prefers the allocation B_H to B_L . But we can also observe that the seller could do even better, by offering the same allocation to the θ_L -type buyer, and some other allocation $B'_H \neq B_H$ to the θ_H -type buyer: At B'_H , the seller gets a higher transfer $T(q)$ for the same consumption, and the θ_L -type buyer is now indifferent between B'_H and B_L . This observation raises the question of the form of the optimal nonlinear pricing contract.

1.3 The second-best outcome: optimal nonlinear pricing

We show in this section that the seller can improve her payoff by offering more general nonlinear contracts than a single two-part tariff. Since the seller does not observe the buyer's type (here, his willingness to pay for the good), she is forced to offer him a set of choices independent of his type. Without loss of generality, this set can be described as $[q, T(q)]$: the buyer faces a menu from which he will pick the outcome that maximizes his payoff.

- *Step 1: Apply the revelation principle.*

From the revelation principle (see previous lecture note), we know we can restrict our attention to a menu of contracts $\{(T(q_L), q_L), (T(q_H), q_H)\}$ that correspond to the optimal choices made by the two types of buyers. If we define $T(q_i) = T_i$ for $i = L, H$, the principal's problem can be rewritten as

$$\begin{aligned} \max_{T_L, q_L, T_H, q_H} \quad & \beta[T_L - cq_L] + (1 - \beta)[T_H - cq_H] \\ \text{subject to} \quad & \text{IC}_L : \theta_L v(q_L) - T_L \geq \theta_L v(q_H) - T_H \\ & \text{IC}_H : \theta_H v(q_H) - T_H \geq \theta_H v(q_L) - T_L \\ & \text{IR}_L : \theta_L v(q_L) - T_L \geq 0 \\ & \text{IR}_H : \theta_H v(q_H) - T_H \geq 0 \end{aligned}$$

- *Step 2: Observe that IR_H is implied by IR_L and IC_H .*

Indeed, $\theta_H v(q_H) - T_H \geq \theta_H v(q_L) - T_L \geq \theta_L v(q_L) - T_L \geq 0$, where the second inequality simply comes from the fact that $\theta_H > \theta_L$.

Note that IR_H will not bind at the optimum.

- *Step 3: IR_L must bind.*

Suppose it did not. Then, $\theta_H v(q_H) - T_H \geq \theta_H v(q_L) - T_L \geq \theta_L v(q_L) - T_L > 0$, and the Principal could increase T_L and T_H by the same amount, which would strictly increase her payoff, without affecting incentive compatibility. Hence in optimum $\theta_L v(q_L) - T_L = 0$

- *Step 4: IC_H must bind.*

Suppose it did not. Then $\theta_H v(q_H) - T_H > \theta_H v(q_L) - T_L > \theta_L v(q_L) - T_L = 0$,

and the Principal could augment T_H without breaking incentive compatibility or the participation constraint of type- θ_L buyer, IR_L . Hence in optimum $\theta_H v(q_H) - T_H = \theta_H v(q_L) - T_L$.

- *Step 5: $q_H \geq q_L$.*
Rewrite IC_L and IC_H .

$$\begin{aligned}\theta_L (v(q_L) - v(q_H)) - T_L + T_H &\geq 0 \\ \theta_H (v(q_H) - v(q_L)) - T_H + T_L &\geq 0\end{aligned}$$

Subtract the two inequalities. The transfers T_i cancel out, and we obtain $\theta_H(v(q_H) - v(q_L)) \geq \theta_L(v(q_H) - v(q_L))$. Since $\theta_H > \theta_L$ and $v' > 0$, it must be that $q_H \geq q_L$.

- *Step 6: IC_L is satisfied at the optimum.*

Using the fact that IC_H binds, we can write $\theta_H(v(q_H) - v(q_L)) = T_H - T_L$. Since $\theta_H > \theta_L$, $\theta_L(v(q_H) - v(q_L)) < T_H - T_L$, which can be rewritten as IC_L : $\theta_L v(q_L) - T_L > \theta_L v(q_H) - T_H$.

Step 7: Solve the maximization problem by substituting the two remaining binding constraints in the principal's objective function.

To summarize, IR_L and IC_H are binding, and we can neglect IR_H and IC_L . Using these properties, we can write IR_L as $T_L = \theta_L v(q_L)$ and, after substituting into IC_H , $T_H = \theta_H v(q_H) - (\theta_H - \theta_L)v(q_L)$. The principal's optimization problem is thus

$$\max_{q_L, q_H} \beta[\theta_L v(q_L) - cq_L] + (1 - \beta)[\theta_H v(q_H) - cq_H - (\theta_H - \theta_L)v(q_L)]. \quad (10)$$

Note that we have eliminated the T_i 's from the problem as these are determined by the binding constraints (IR_L and IC_H). Instead of 4 variables to solve for and 4 lagrange multipliers we are left with 2 first order conditions!

The first term in brackets is the full surplus generated by the purchases from the type- θ_L buyer, which the principal appropriates entirely. The second term in brackets is the full surplus generated by purchases from the type- θ_H buyer minus his *informational rent* $(\theta_H - \theta_L)v(q_L)$. This term represents the amount of surplus the principal must leave to the type- θ_H buyer, if she does not want him to mimic the behavior of the other type. Note that this informational rent is increasing in q_L . This also illustrates the important trade-off the principal faces. By decreasing q_L the principal decreases the informational rent "paid" to the high type, but forgoes trade with the low type.

The FOCs characterize the unique interior solution (q_L^{SB}, q_H^{SB}) , if this solu-

tion exists:¹

$$\theta_H v'(q_H^{SB}) = c, \quad (11)$$

$$\theta_L v'(q_L^{SB}) = \frac{c}{1 - \left(\frac{1-\beta}{\beta} \frac{\theta_H - \theta_L}{\theta_L} \right)} > c. \quad (12)$$

We immediately confirm that this solution implies $q_L^{SB} < q_H^{SB}$, and we can immediately check that the two omitted constraints IC_L and IR_H are indeed verified at the optimum $(q_i^{SB}, T_i^{SB})_{i=L,H}$ given that IC_H binds. Indeed, $\theta_H v(q_H^{SB}) - T_H^{SB} = \theta_H v(q_L^{SB}) - T_L^{SB}$, together with $\theta_L < \theta_H$ and $q_L^{SB} < q_H^{SB}$, implies $\theta_L v(q_H^{SB}) - T_H^{SB} \leq \theta_L v(q_L^{SB}) - T_L^{SB} : IC_L$.

Two basic economic conclusions emerge from this analysis:

1. The second-best optimal consumption for the type- θ_H buyer is the same as the first-best optimal consumption q_H^{FB} . Hence the expression “No distortion at the top”: the allocation of the “good” type is not distorted with respect to the efficient allocation (more generally, the allocation of the type who would like to mimic the other type is not distorted at the second-best optimum). Conversely, the consumption of the type θ_L is lower.
2. The type- θ_L buyer obtains a surplus of zero, while the other type obtains strictly positive “informational” rent.

The consumption distortion for type θ_L is the result of the Principal’s attempt to reduce the informational rent she must leave to type θ_H . Since a buyer of type θ_H is more eager to consume, the seller can reduce that type’s incentive to mimic the type θ_L by cutting down on the consumption offered to type θ_L . By reducing type θ_H ’s incentives to mimic type θ_L , the seller can reduce the informational rent (or equivalently, charge a higher price) of type θ_H .

Looking at the FOC for q_L^{SB} shows that the size of the distortion $q_L^{FB} - q_L^{SB}$ is increasing in the potential size of the informational rent, as measured by $(\theta_H - \theta_L)$, and decreasing in β .

¹If the denominator of the second expression is not positive, then the optimal solution involves $q_L^{SB} = 0$, while q_H^{SB} remains determined by the first expression.